

PhD. Models and Their Sub Models – Documentation

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Wisconsin Models PDF

Ph.D. Module Overview

The **Ph.D. Module** is designed to manage the complete academic and research journey of a doctoral student within the university. It covers every stage of the Ph.D. lifecycle, beginning from admission into a doctoral program and continuing until the student successfully graduates. The module helps the university organize student information, research activities, faculty supervision, coursework, examinations, dissertation work, publications, annual evaluations, and graduation records in a structured manner.

The primary objective of this module is to ensure that every Ph.D. student's academic progress is properly monitored while maintaining transparency and consistency throughout the research process.

Main Model

1. Ph.D. Program Model

Purpose

The **Ph.D. Program Model** acts as the parent model of the entire Ph.D. module. It stores the basic information about every doctoral program offered by a department. Every Ph.D. student admitted into the university will be associated with one of these programs.

Fields

- phd_program_id
- department_id (FK)
- program_name
- degree_type (PhD)
- total_credits_required
- residency_requirement

- duration_years
- program_description
- status (Active / Inactive)

Description

This model defines important academic requirements such as the total credits a student must complete, the minimum residency period, program duration, and the current status of the program. Since all other Ph.D.-related records depend on this model, it serves as the foundation of the entire doctoral management system.

Sub Models

2. Ph.D. Student Model

Purpose

The **Ph.D. Student Model** stores the complete academic profile of every doctoral student enrolled in the university.

Fields

- phd_student_id
- student_id
- phd_program_id
- advisor_id
- admission_date
- cohort_year
- current_status
- expected_graduation_date

Description

This model links a student with a specific Ph.D. program and research advisor. It also keeps track of admission details, the student's current academic status, and the expected completion date of the degree.

Relationship

One Ph.D. Program → Many Ph.D. Students

3. Research Advisor Model

Purpose

The **Research Advisor Model** stores information about faculty members who guide and supervise doctoral research.

Fields

- advisor_id
- faculty_id
- research_area
- lab_name
- available_slots
- publications_count

Description

Every Ph.D. student is assigned a research advisor who provides academic guidance throughout the research journey. This model records the advisor's specialization, laboratory details, publication count, and the number of students they can supervise.

Relationship

One Research Advisor → **Many Ph.D. Students**

4. Doctoral Committee Model

Purpose

The **Doctoral Committee Model** maintains the committee formed to evaluate and monitor a student's research progress.

Fields

- committee_id
- phd_student_id
- chair_faculty_id
- formation_date
- approval_status

Description

A doctoral committee is responsible for reviewing research progress, evaluating proposals, conducting examinations, and approving major academic milestones. This model stores committee formation details and approval information.

Relationship

One Ph.D. Student → One Doctoral Committee

5. Committee Member Model

Purpose

The **Committee Member Model** stores all faculty members who are part of a doctoral committee.

Fields

- member_id
- committee_id
- faculty_id
- role
- department

Committee Roles

- Chair
- Co-Chair
- Internal Member
- External Member

Description

Each committee may consist of multiple faculty members from different departments. Every member performs a specific responsibility during proposal reviews, examinations, dissertation evaluations, and defense sessions.

Relationship

One Doctoral Committee → Many Committee Members

6. Coursework Requirement Model

Purpose

The **Coursework Requirement Model** records all coursework completed by a Ph.D. student before entering the research phase.

Fields

- coursework_id
- phd_student_id
- course_id
- credits
- grade
- completion_status

Description

Many doctoral programs require students to complete mandatory coursework. This model stores completed courses, earned credits, grades, and overall completion status.

Relationship

One Student → Many Coursework Records

7. Qualifying / Preliminary Examination Model (Exam conducted after completing the required coursework and before starting independent research.)

Purpose

The **Qualifying / Preliminary Examination Model** stores information related to qualifying examinations conducted before dissertation (a long piece of writing on something that you have studied, especially as part of a university degree. ஆராய்ந்த ஒன்று குறித்த நீண்ட எழுத்துரை. குறிப்பாகப் பல்கலைக்கழகப் பட்டத்துக்கான பகுதிப்பணியான நீண்ட ஆய்வுக்கட்டுரை. ஆய்வேடு.) research begins.

Fields

- prelim_exam_id
- phd_student_id
- exam_date

- exam_type
- result
- remarks

Description

Students must successfully clear qualifying examinations before proceeding to independent research. This model records examination details, results, and committee remarks.

Relationship

One Student → Multiple Examination Records (in case first exam fail aaita)

8. Dissertation Proposal Model (athavathu official ah university ku submit pandra works like documents which he created to submit in the university of what he going to do)

Purpose

The **Dissertation Proposal Model** stores information about the student's proposed research topic.

Fields

- proposal_id
- phd_student_id
- proposal_title
- abstract
- submission_date
- hearing_date
- result
- approved_by (**doctoral committee ku dhan andha authority iruku research advisor ku kedaiyadhu**)

Description

Before beginning research, students submit a dissertation proposal explaining the research objectives, methodology, and expected outcomes. This model stores proposal details and approval information.

Relationship

One Student → One Dissertation Proposal (in rare cases if the proposal gets rejected the universities may allow the student to change the topic)

9. Doctoral Candidacy Model (oru student oru researcher aaga eligible aaitan or he may proceed to his own independent research ngra maari)

Purpose

The **Doctoral Candidacy Model** records whether a student has officially advanced to doctoral candidacy.

Fields

- candidacy_id
- phd_student_id
- candidacy_date
- candidacy_status
- approval_date

Description

After completing coursework, passing qualifying examinations, and obtaining proposal approval, students become official doctoral candidates. This model records that transition.

Relationship

One Student → One Doctoral Candidacy Record

10. Ph.D. Milestone Model

Purpose

The **Ph.D. Milestone Model** tracks important academic milestones completed during the research period.

Fields

- milestone_id
- phd_student_id
- milestone_name
- completion_date
- status

Description

Every Ph.D. student completes multiple milestones throughout the program, such as proposal approval, research completion, publication submission, and dissertation defense. This model records the completion status of each milestone.

Relationship

One Student → Multiple Milestone Records

11. Dissertation Model (total documents which he created throughout his journey from literally everything)

Purpose

The **Dissertation Model** stores the final research work submitted by the student.

Fields

- dissertation_id
- phd_student_id
- dissertation_title
- abstract
- submission_date
- current_version
- status

Description

The dissertation represents the student's original research contribution. This model stores dissertation details, submission information, version history, and approval status.

Relationship

One Student → One Dissertation

12. Research Publication Model

Purpose

The **Research Publication Model** stores publications produced by Ph.D. students during their research.

Fields

- publication_id
- phd_student_id
- title
- journal
- publication_date
- DOI
- indexed_status

Description

Research publications demonstrate the academic contribution of a doctoral student. This model records journal publications, DOI information, publication dates, and indexing status.

Relationship

One Student → Multiple Research Publications

13. Annual Progress Review Model (indha varsham mudinchu nee enna paniruka nu kekra maari like gathering and making all info in a documents in an annual year. Mainly this if for the committee to check the progress of the PhD student)

Purpose

The **Annual Progress Review Model** records yearly evaluations of a student's research performance.

Fields

- review_id
- phd_student_id
- review_year
- committee_comments
- progress_score
- status

Description

Every academic year, the doctoral committee reviews the student's research progress. This model stores committee observations, progress scores, recommendations, and overall review status.

Relationship

One Student → One Annual Progress Review per Year

14. Dissertation Defense Model (Like Viva)

Purpose

The **Dissertation Defense Model** stores information related to the final dissertation defense.

Fields

- defense_id
- phd_student_id
- defense_date
- location
- result
- committee_decision
- final_comments

Description

Once the dissertation is completed, students present and defend their research before the doctoral committee. This model stores defense results, committee decisions, and final remarks.

Relationship

One Student → One Dissertation Defense Record

15. Graduation Model

Purpose

The **Graduation Model** stores the final academic records after successful completion of the Ph.D. program.

Fields

- graduation_id
- phd_student_id
- graduation_date
- degree_awarded

- final_gpa
- dissertation_accepted

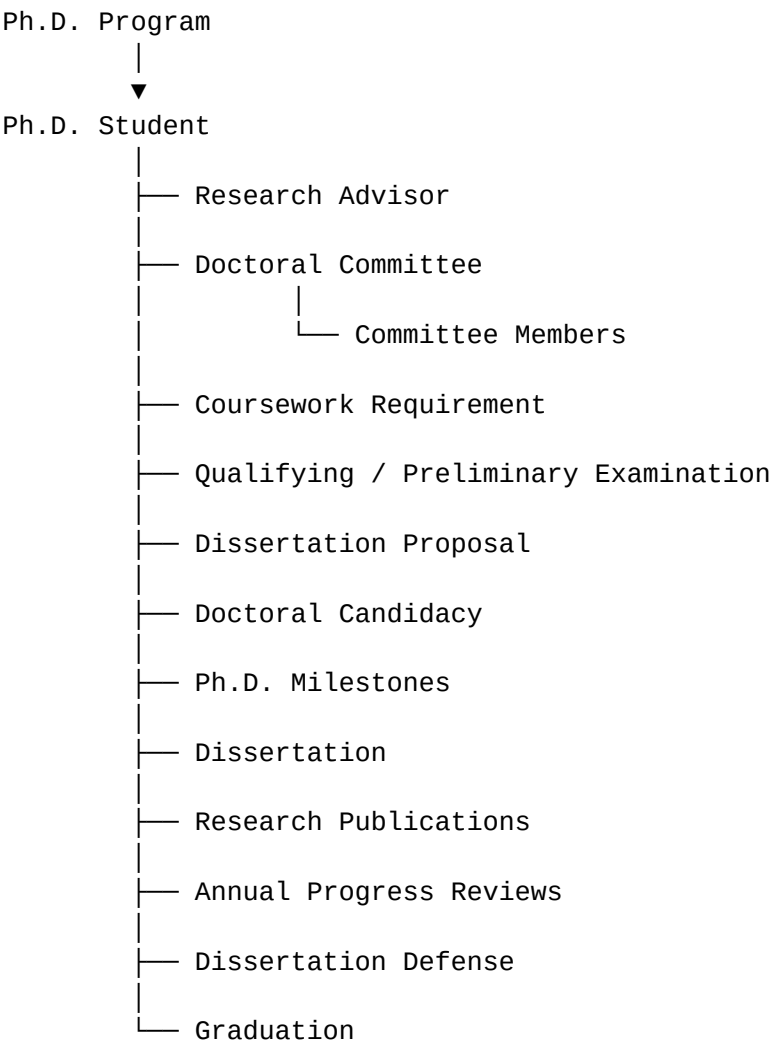
Description

This is the final stage of the doctoral journey. The model records graduation details, confirms dissertation acceptance, stores the final GPA, and indicates whether the doctoral degree has been officially awarded.

Relationship

One Student → One Graduation Record

Overall Workflow



Summary

The **Ph.D. Module** consists of **one main model (Ph.D. Program Model)** and **fourteen supporting sub-models**, each representing a critical stage in a doctoral student's academic and research journey. Together, these models create a comprehensive system that enables universities to manage doctoral education efficiently—from program creation and student admission to research supervision, committee evaluations, coursework completion, qualifying examinations, dissertation development, publication tracking, annual progress assessments, dissertation defense, and final graduation. This structured design ensures accurate record management, better academic monitoring, and a well-organized workflow throughout the entire Ph.D. lifecycle.